# Chapter 8 Review

*Campbell Biology, AP Edition, 12e*

# Chapter 8 Review

An image depicting a phenomenon known as bioluminescence, where organisms convert chemical energy to light energy. Pyrophorus nyctophanus, the click beetle, uses bioluminescence in its larval stages to attract termites from the termite mound for its food.

## Summary of Key Concepts

Virtually all aerobic organisms share some similar enzymes, such as those involved in carbohydrate metabolism and in hydrogen peroxide degradation. How is it possible to have specific shared functions in enzymes when organisms and their internal and external environments vary so much? **(BIG IDEAS 1 & 4)**

### Concept 8.1: An organism's metabolism transforms matter and energy

- ****Metabolism** is the collection of chemical reactions that occur in an organism. Enzymes catalyze reactions in intersecting **metabolic pathways**, which may be **catabolic** (breaking down molecules, releasing energy) or **anabolic** (building molecules, consuming energy). **Bioenergetics** is the study of the flow of energy through living organisms.
- ****Energy** is the capacity to cause change; some forms of energy do work by moving matter. **Kinetic energy** is associated with motion and includes **thermal energy** associated with random motion of atoms or molecules. **Heat** is thermal energy in transfer from one object to another. **Potential energy** is related to the location or structure of matter and includes **chemical energy** possessed by a molecule due to its structure.
- The **first law of thermodynamics**, conservation of energy, states that energy cannot be created or destroyed, only transferred or transformed. The **second law of thermodynamics** states that ****spontaneous processes**, those requiring no outside input of energy, increase the **entropy** (molecular disorder) of the universe.

*Explain how the highly ordered structure of a cell does not conflict with the second law of thermodynamics.*

## Concept 8.2: The free-energy change of a reaction tells us whether or not the reaction occurs spontaneously

- A living system's **free energy** is energy that can do work under cellular conditions. The change in free energy ($\Delta G$) during a biological process is related directly to enthalpy change ($\Delta H$) and to the change in entropy ($\Delta S$):$\Delta G = \Delta H - T\Delta S$. Organisms live at the expense of free energy. A spontaneous process occurs with no energy input; during such a process, free energy decreases and the stability of a system increases. At maximum stability, the system is at equilibrium and can do no work.
- In an **exergonic** (spontaneous) chemical reaction, the products have less free energy than the reactants ($-\Delta G$). **Endergonic** (nonspontaneous) reactions require an input of energy ($+\Delta G$). The addition of starting materials and the removal of end products prevent metabolism from reaching equilibrium.

*Explain the meaning of each component in the equation for the change in free energy of a spontaneous chemical reaction. Why are spontaneous reactions important in the metabolism of a cell?*

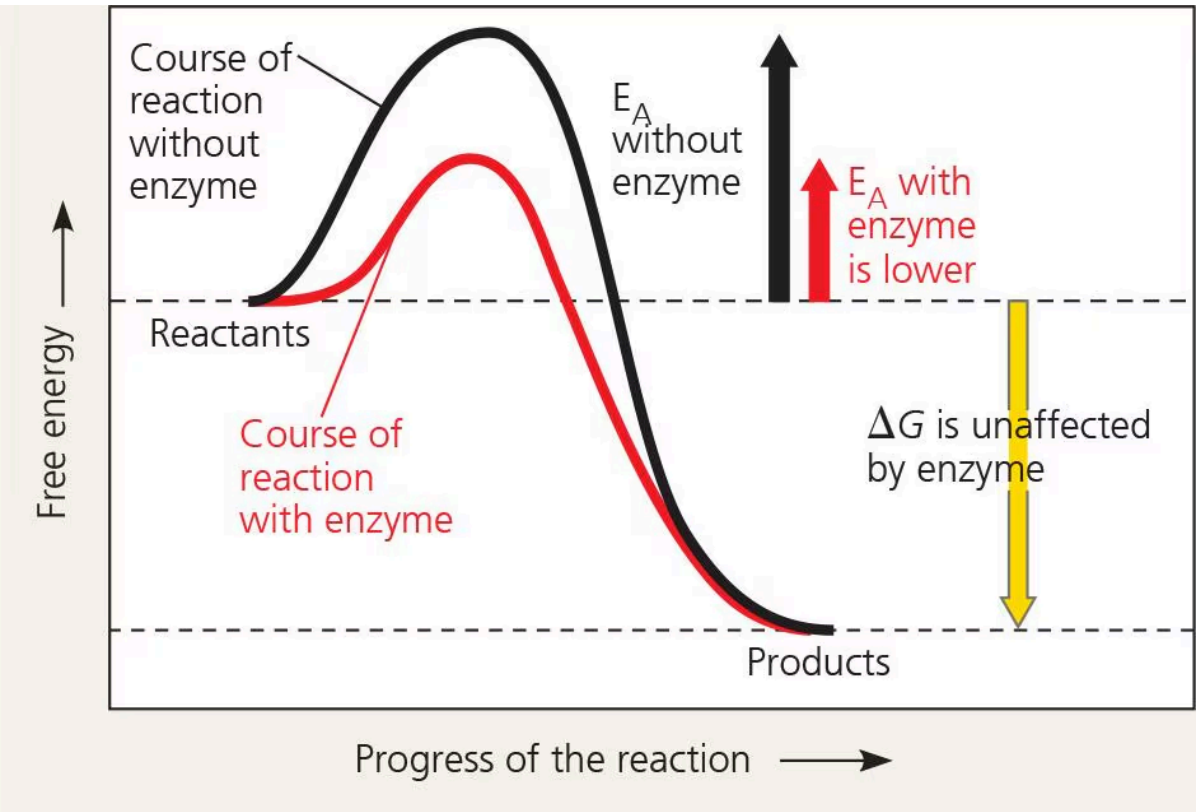
## Concept 8.3: ATP powers cellular work by coupling exergonic reactions to endergonic reactions

- **ATP** is the cell's energy shuttle. Hydrolysis of its terminal phosphate yields ADP and Ⓟi and releases free energy.
- Through ****__energy coupling__, the exergonic process of ATP hydrolysis drives endergonic reactions by transfer of a phosphate group to specific reactants, forming a **phosphorylated intermediate** that is more reactive. ATP hydrolysis (sometimes with protein phosphorylation) also causes changes in the shape and binding affinities of transport and motor proteins.
- Catabolic pathways drive regeneration of ATP from ADP+Ⓟi.

*Describe the ATP cycle: How is ATP used and regenerated in a cell?*

## Concept 8.4: Enzymes speed up metabolic reactions by lowering energy barriers

- In a chemical reaction, the energy necessary to break the bonds of the reactants is the ****__activation energy__, EA.
- ****__Enzymes__ lower the EA barrier:




Free energy is on the y-axis and Progress of the reaction is on the x-axis. E sub A without enzyme is shown to be approximately twice as large as E sub A with enzyme. Delta G is unaffected by enzyme.

- Each enzyme has a unique **active site** that binds one or more **substrate(s)**, the reactants on which it acts. It then changes shape, binding the substrate(s) more tightly (**induced fit**).
- The active site can lower an EA barrier by orienting substrates correctly, straining their bonds, providing a favorable microenvironment, or even covalently bonding with the substrate.
- Each enzyme has an optimal temperature and pH.
- Inhibitors reduce enzyme function. A **competitive inhibitor** binds to the active site, whereas a **noncompetitive inhibitor** binds to a different site on the enzyme.
- Natural selection, acting on organisms with variant enzymes, is responsible for the diversity of enzymes found in organisms.

*How do both activation energy barriers and enzymes help maintain the structural and metabolic order of life?*

## Concept 8.5: Regulation of enzyme activity helps control metabolism

- Many enzymes are subject to **allosteric regulation**: Regulatory molecules, either activators or inhibitors, bind to specific regulatory sites, affecting the shape and function of the enzyme. In **cooperativity**, binding of one substrate molecule can stimulate binding or activity at other active

sites. In **feedback inhibition**, the end product of a metabolic pathway allosterically inhibits the enzyme for a previous step in the pathway.

- Some enzymes are grouped into complexes, some are incorporated into membranes, and some are contained inside organelles, increasing the efficiency of metabolic processes.

*What roles do allosteric regulation and feedback inhibition play in the metabolism of a cell?*

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